

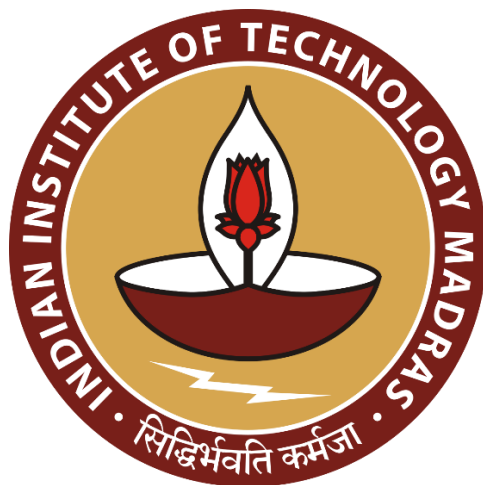
Multimodal Misinformation Detector Agent

Data Science and AI Lab Project

Milestone 2 Report

Group 9

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1. Objective

Under milestone 2, data understanding and preprocessing were performed on both image and text-based misinformation detection pipelines. Extensive EDA was conducted on both modalities to:

- Assess class balance (e.g., fake vs. real, truthfulness labels).
- Identify potential quality issues, duplicates, or corrupted samples.
- Standardize data formats (image resolution, encoding, labeling etc.).
- Prepare consistent train, validation, and test splits for reproducibility.

2. Datasets Overview

2.1 Image Dataset

- **Source** – [Kaggle – AI-Generated Images vs Real Images](#)
- **Description** – This dataset presents a balanced and diverse collection of 60,000 labeled images, equally divided between *AI-generated* and *real* categories, ensuring fair representation for supervised model training. The data spans a wide range of visual domains including landscapes, people, objects, and abstract art, enabling the model to generalize beyond face-centric or narrow-domain forgeries.
- **Rationale for choice:**
 - Comprehensive coverage of modern generative models
 - Images from multiple AI generation systems capturing stylistic diversity
 - Natural and artistic real images mitigating overfitting risks
 - Publicly available with an MIT license
 - Sufficient quantity for training, validation, and testing
 - Suitable for binary classification with balanced class structure

2.2 Text Dataset - 1

- **Source** – [HuggingFace – Fever Dataset](#)

- **Description** – The **FEVER (Fact Extraction and VERification)** dataset, released by the University of Sheffield and the University of Washington, is a large-scale benchmark designed for automated fact verification tasks. It contains approximately **185,445** human-generated claims, each annotated as *SUPPORTED*, *REFUTED*, or *NOT ENOUGH INFO*, based on evidence retrieved from Wikipedia. For each claim, relevant evidence sentences are also provided, enabling both classification and retrieval-based evaluation.
- **Rationale for choice:**
 - Covers multiple domains and claim types
 - High-quality human-annotated evidence and labels
 - Supports multi-hop reasoning and evidence retrieval
 - Suitable for fine-tuning LLMs in credibility analysis
 - Available with GNU General Public License v3.0
- **References:** Thorne, J., Vlachos, A., Christodoulopoulos, C., & Mittal, A. (2018). FEVER: A large-scale dataset for fact extraction and verification. In *Proceedings of NAACL-HLT*.

2.3 Text Dataset - 2

- **Source** – [Kaggle – LIAR Dataset](#)
- **Description** – The **LIAR dataset** is a benchmark dataset for fake news detection and credibility analysis, introduced by William Yang Wang in 2017. It contains over 12,000 short statements labeled across six truthfulness categories, ranging from “pants fire” to “true,” along with rich metadata including the speaker, job title, party affiliation, state, context, and historical credibility counts. The dataset is structured in a TSV format. Its comprehensive labeling and contextual information make it a valuable resource for training models that evaluate statement veracity.
- **Rationale for choice:**
 - Richly labeled for truthfulness evaluation

- Captures historical credibility
- Short statements suitable for rapid model training
- Publicly available with Apache 2.0 license and research-friendly

3. Exploratory Data Analysis (EDA)

3.1 Image Dataset

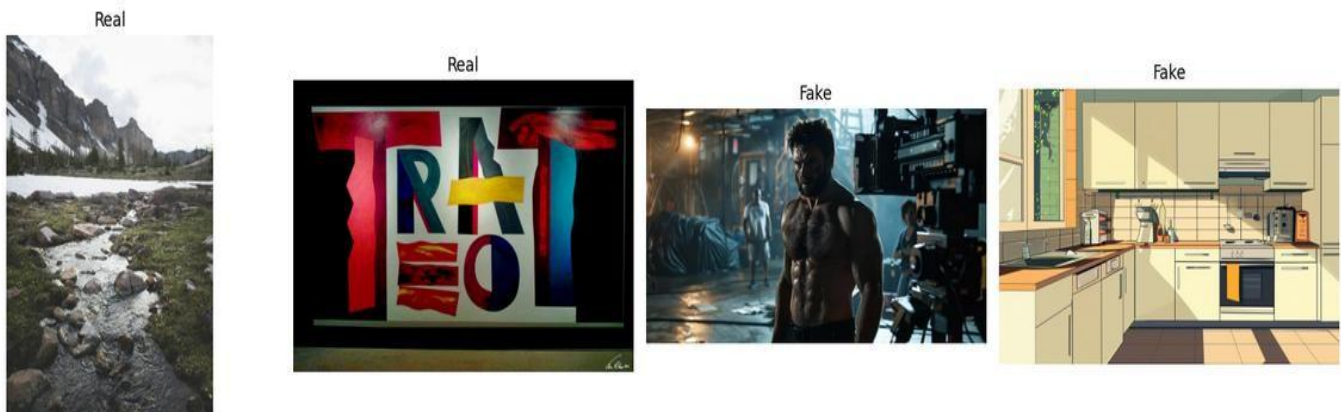
The image dataset consisted of AI-generated and real images. EDA was performed to understand class distributions, visual characteristics, and preprocessing requirements.

- **Total Dataset Size and Split:**

The dataset contains a total of 60,000 images, with 30,000 AI-generated and 30,000 real images. For model training, the dataset was split into training (64%), validation (16%), and testing (20%) sets, with equal representation from real and fake images.

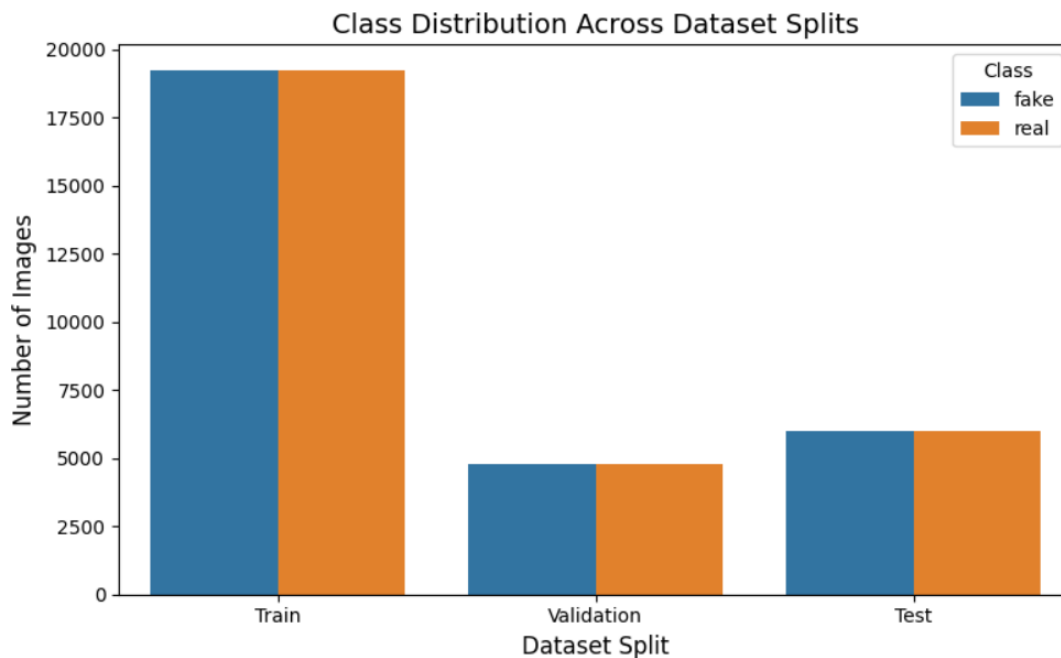
- **Sample Images:**

Random samples were visualized to observe the differences between AI-generated and real images.



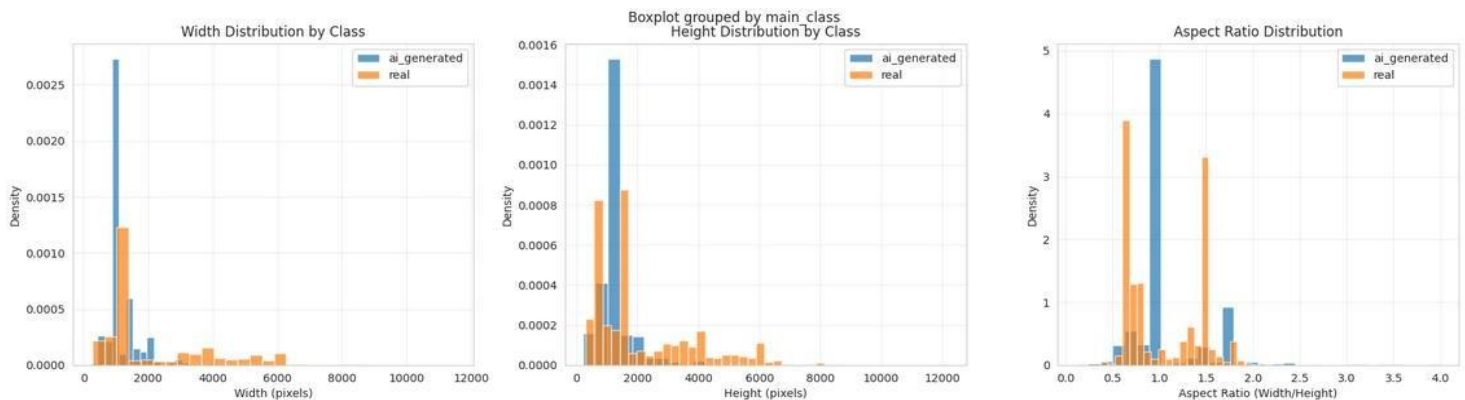
- **Class Distribution:**

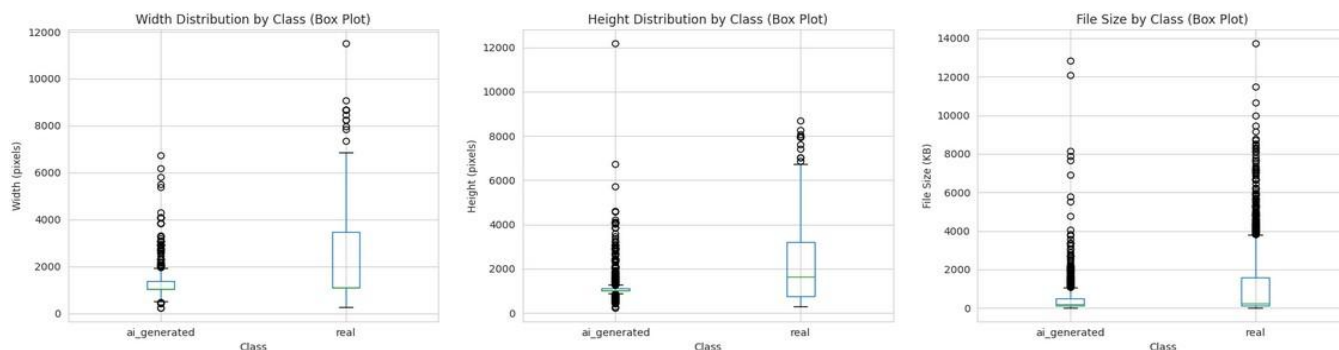
The dataset is balanced across classes, which reduces potential bias in model training.



- **Image Resolution and Size:**

The images vary in resolution, with both width and height peaking around 1000 pixels. The distribution is right-skewed, indicating that while most images are around 1000×1000 pixels, there are some high-resolution outliers. Preprocessing such as resizing and normalization was done to ensure consistency across the dataset.

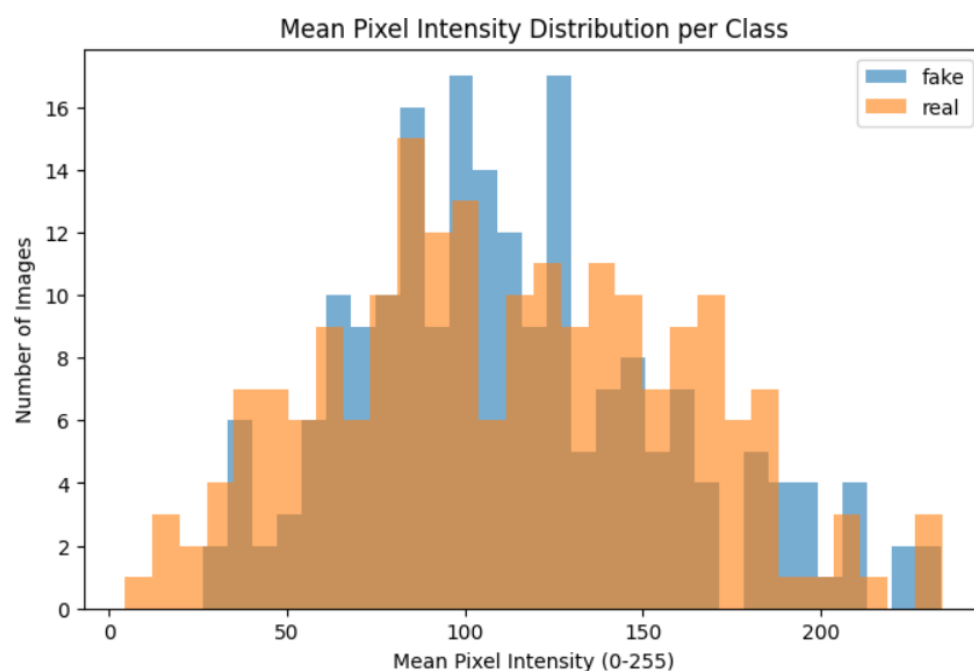




Both AI-generated and real images have similar aspect ratios (≈ 1.1 average), mostly near square. Real images are larger on average (1,142.6 KB) than AI-generated (442.5 KB), reflecting higher resolution and more detail.

- **Color and Pixel Intensity Analysis:**

The mean pixel intensity distributions for both real and AI-generated images resemble a single-peaked (hill-shaped) curve, with peaks occurring at slightly different but mostly random intensity values. Overall, the distributions have a similar shape.



- **Augmentation and Transformations:**

Sample augmentations (rotation, flipping, brightness adjustment) were applied to simulate real-world variability and improve model robustness.



To summarize, the EDA revealed that the dataset is balanced and ready for modeling, with preprocessing needed for consistent image size and normalization. Augmentations will help improve model generalization during training.

3.2 Text Dataset - 1

The dataset contains claims labeled as SUPPORTS, REFUTES, or NOT ENOUGH INFO, along with evidence sentences from Wikipedia. EDA was performed to understand the dataset's structure, label distribution, claim characteristics, and evidence information.

- **Data Overview:**

The dataset contains 145,449 claims with 5 columns.

Column	Description
id	Unique identifier for each claim
verifiable	Indicates whether sufficient evidence exists for verification (VERIFIABLE / NOT VERIFIABLE).
label	The verification outcome (SUPPORTS / REFUTES / NOT ENOUGH INFO)
claim	The textual statement whose truthfulness is verified
evidence	A list of lists. Each inner list is an annotation group. [annotator_id, evidence_id, Wikipedia_title, sentence_index]

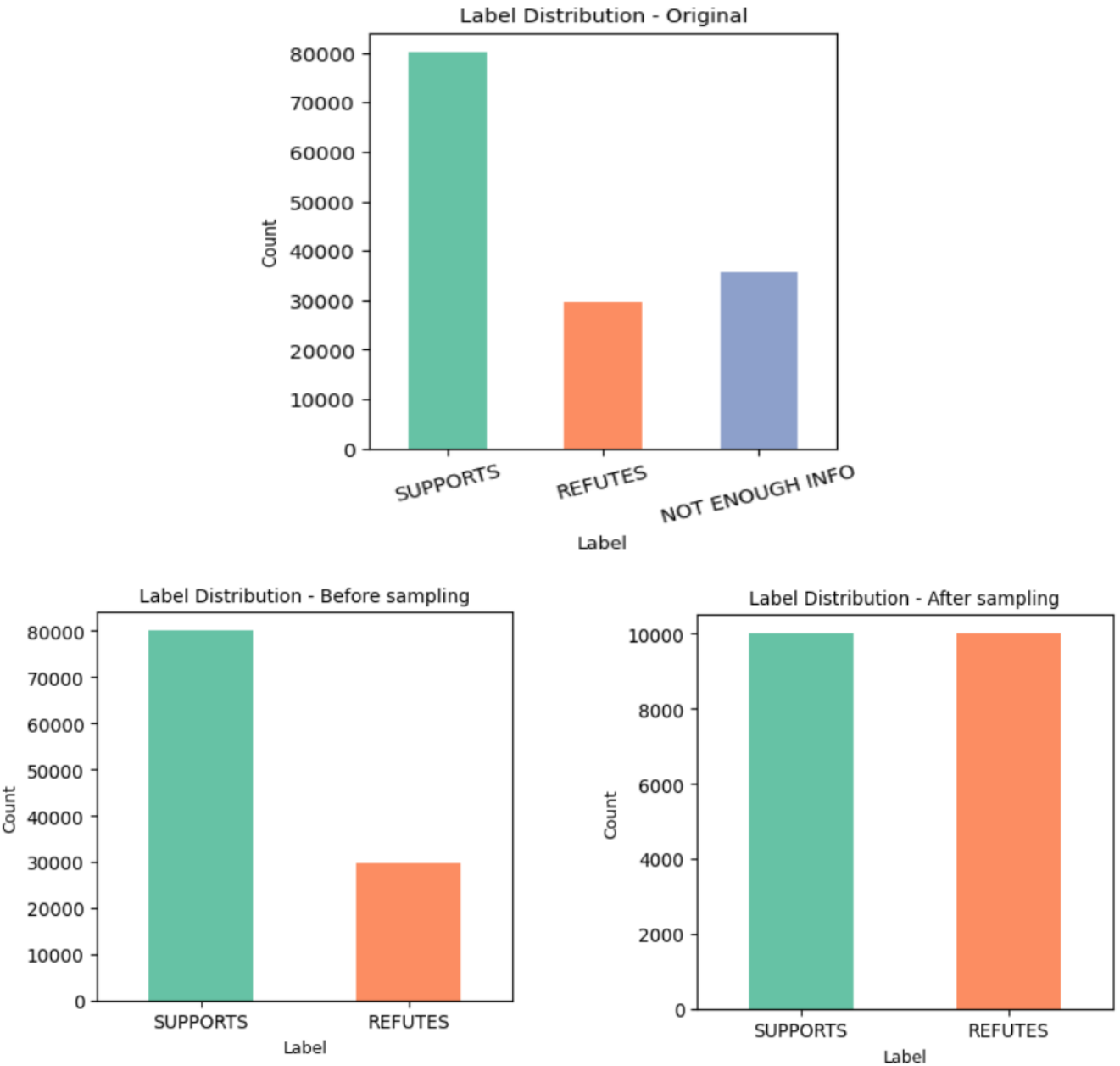
- **Sample Data**

Below are top 5 rows from the dataset spanning across different labels, different values of verifiability.

id	verifiable	label	claim	evidence
75397	VERIFIABLE	SUPPORTS	Nikolaj Coster-Waldau worked with the Fox Broa...	[[[92206, 104971, Nikolaj_Coster-Waldau, 7], [...
150448	VERIFIABLE	SUPPORTS	Roman Atwood is a content creator.	[[[174271, 187498, Roman_Atwood, 1]], [[174271...
214861	VERIFIABLE	SUPPORTS	History of art includes architecture, dance, s...	[[[255136, 254645, History_of_art, 2]]]
156709	VERIFIABLE	REFUTES	Adrienne Bailon is an accountant.	[[[180804, 193183, Adrienne_Bailon, 0]]]
83235	NOT VERIFIABLE	NOT ENOUGH INFO	System of a Down briefly disbanded in limbo.	[[[100277, None, None, None]]]

- Label Distribution – Discard unnecessary data

The dataset contains three labels. However, it is observed that claims with label "NOT ENOUGH INFO" have no evidences. As they do not contribute to training, we could safely drop the corresponding rows.

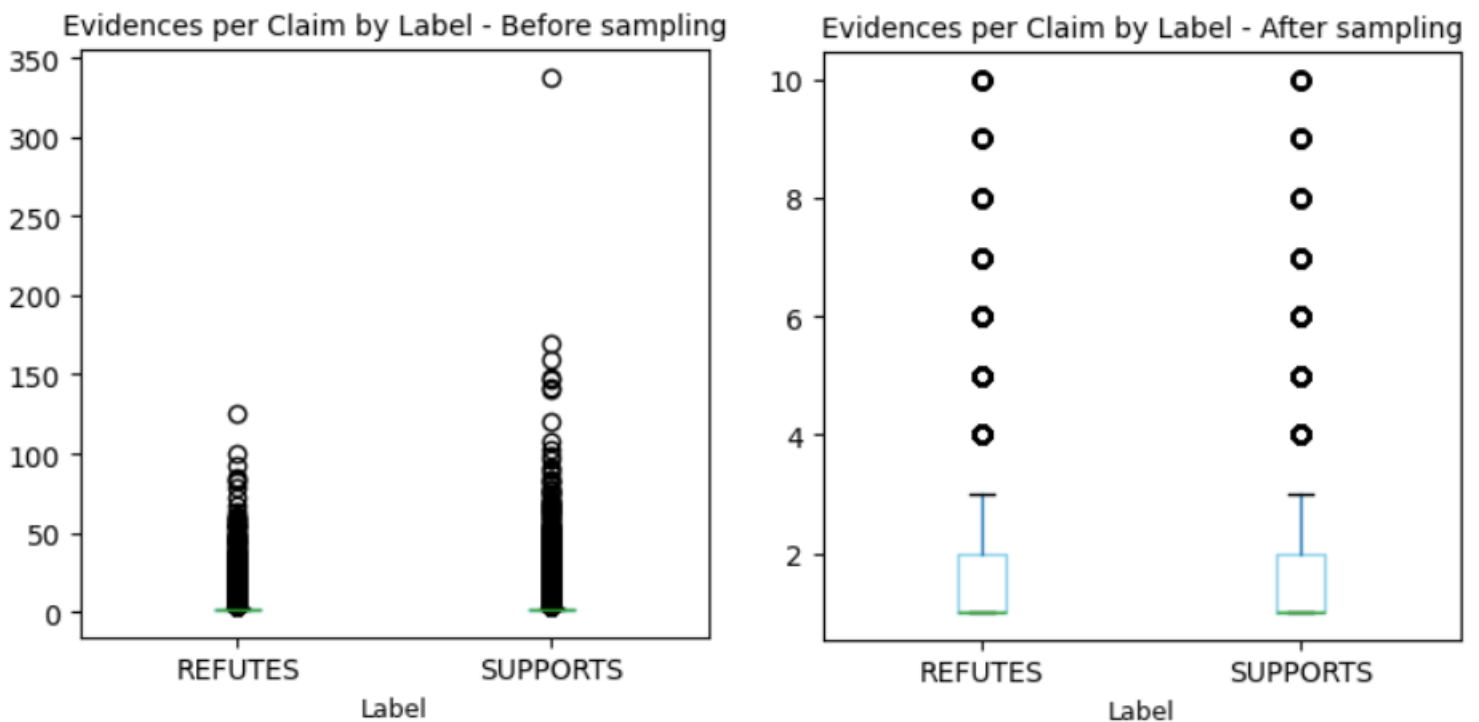


The clear class imbalance, with SUPPORTS being the dominant is also addressed by sampling technique. 10000 claims are sampled thereby a total of 20000 claims.

- **Evidences per claim**

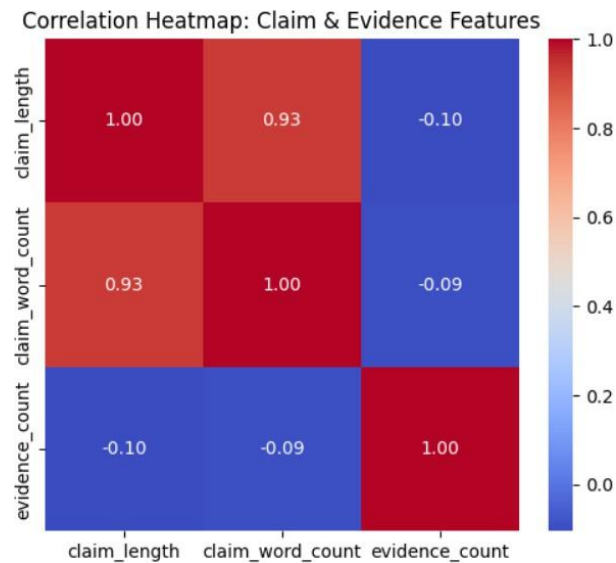
Each claim in the dataset is associated with one or more evidence sentences from Wikipedia. The number of evidence sentences per claim varies widely, ranging from a minimum of 1 to a maximum of 338, with an average of approximately 2.14.

The number of evidences per claim range from 1 to 125 for label REFUTES, and 1 to 338 for label SUPPORTS. Yet, both have a mean around 2. This imbalance is also corrected by sampling claims having evidence counts between 1 and 10.



- **Correlation Analysis**

Claim length and claim word count are highly correlated (0.93), indicating longer claims generally contain more words. Both features show almost no correlation with evidence count (≈ -0.1), suggesting that claim size does not strongly influence the number of evidence sentences.



- **Evidence Text extraction**

The evidence column in the dataset does not directly contain the evidence sentences. A separate wiki-dump, [Pre-processed Wikipedia Pages \(June 2017 dump\)](#) is presented along with the dataset which is to be referenced to extract the sentences. Their evidence texts are extracted accordingly and a new .jsonl file with [claim, evidence_sentences, label] columns is created to be used for training the model.

	claim	evidence_sentences	label
0	Vladimir Putin studied German.	[Born in Leningrad -LRB- now Saint Petersburg ...	SUPPORTS
1	Burundi is a squid.	[Burundi -LRB- -LSB- bæ'rundi -RSB- or -LSB- b...	REFUTES
2	Ronda Rousey's first feature film role was in ...	[Rousey 's first feature film role was the 201...	REFUTES
3	Beautiful People is a film.	[Beautiful People was a British comedy drama t...	REFUTES
4	Richard Nixon defeated George McGovern in 1972.	[He was reelected in one of the largest electo...	SUPPORTS

Overall, the EDA of the FEVER dataset shows a moderate label imbalance, consistent claim lengths, and variable evidence count. The data is clean, structured, and ready for model training with minimal tokenization, and class-balancing steps during training.

3.3 Text Dataset - 2

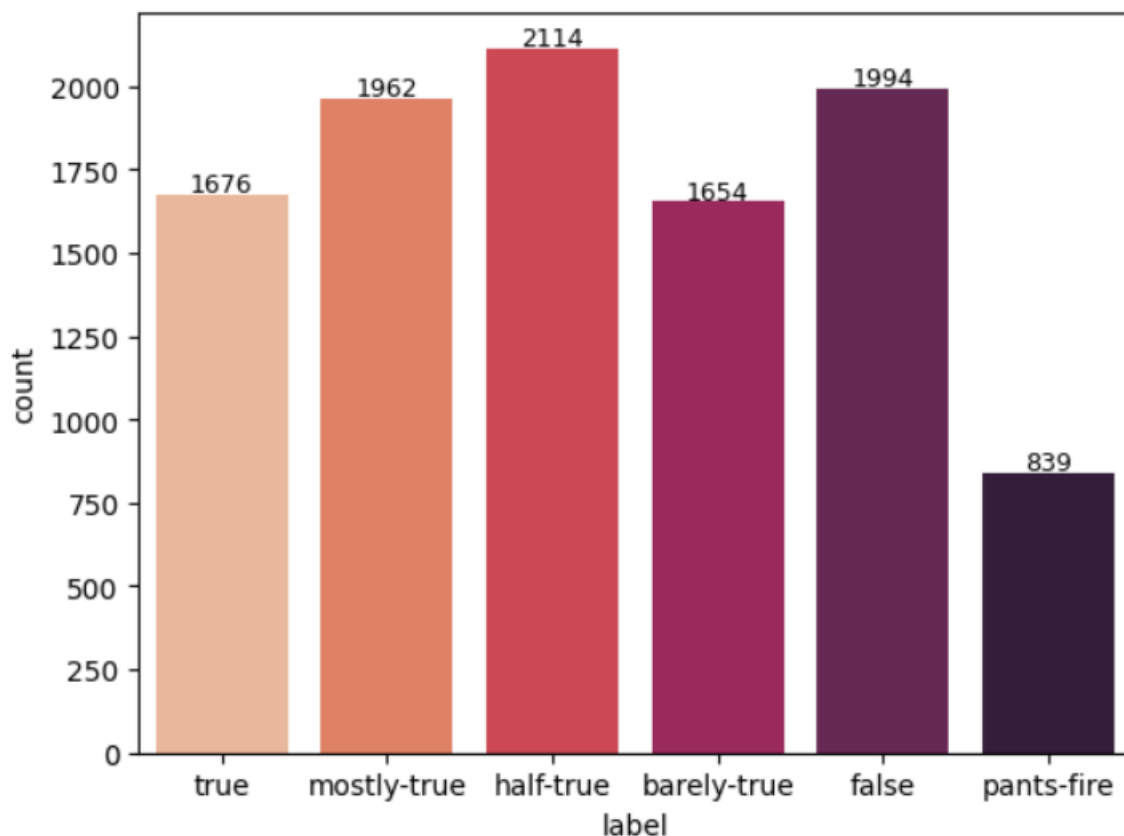
EDA was performed on the LIAR dataset to understand the structure, label distribution, statement characteristics, and metadata patterns. The analysis focused on identifying class imbalances, statement lengths, word counts, and speaker-related features, which are critical for designing effective preprocessing and model training strategies.

- **Data Overview**

The LIAR dataset contains **10,239 short statements** with metadata including speaker, job title, party affiliation, state, subjects, and context. Each statement is labeled for truthfulness across six fine-grained categories.

- **Label Distribution**

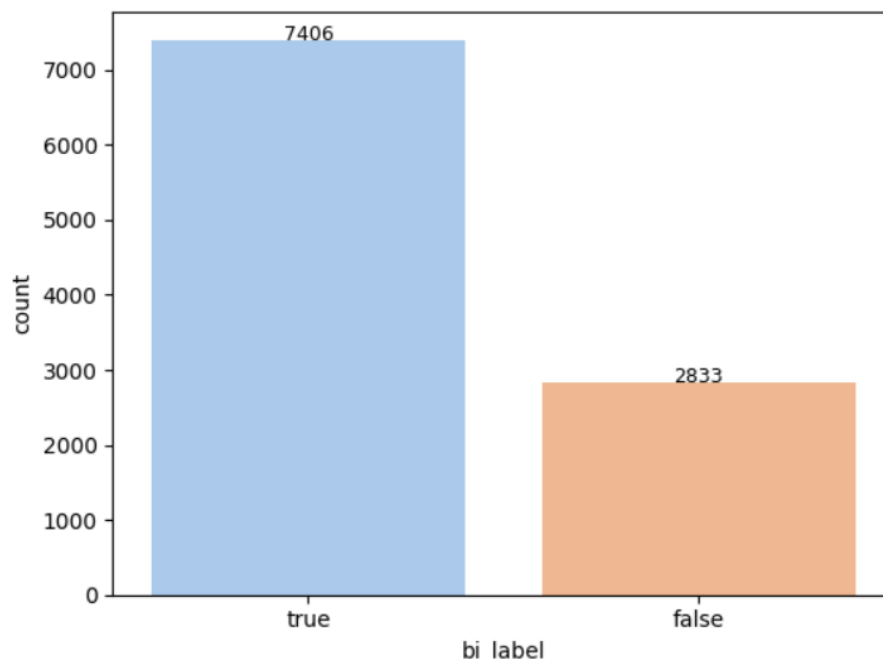
The statements are distributed across six classes as follows.



The statements show a moderate imbalance across the six classes, with pants-fire being the least frequent and False the most common among the negative classes. While training the model, techniques like class weighting or oversampling may be applied to ensure balanced learning across all categories.

- **Binary Label Distribution**

For binary classification, the six labels were consolidated into two classes - statements labeled as true, mostly-true, half-true, or barely-true were mapped to true, while false and pants-fire were mapped to false. The binary label distribution shows a notable imbalance.



Overall, EDA of LIAR dataset shows a moderate class imbalance in the binary labels and relatively uniform statement lengths. The data is structured and ready for model training with class weighting or oversampling techniques.

4. Conclusion

Milestone 2 was successfully completed with the identification, download, and analysis of the datasets. EDA of the text (FEVER, LIAR) and image datasets confirmed that the data is clean, structured, and ready for model training. Label distributions,

statement lengths, evidence counts, and image properties were analyzed, and preprocessing strategies such as truncation, padding, normalization, and class-balancing have been planned.

For Milestone 3, we will fine-tune **DeBERTa** models for textual misinformation detection and use **CLIP** for image classification. Insights from EDA will guide preprocessing, feature selection, and training strategies. These steps lay the groundwork for building a multimodal system combining text and image information for robust misinformation detection.